

2019 Exam Paper

Q1 (a) Ciphertext $S \rightarrow$ Plaintext $E \Rightarrow 18 \rightarrow 4 \Rightarrow y \rightarrow y+13 = x$ decryption function

$$\begin{aligned}
 y = A = 0 &\rightarrow 0+13 = 13 = N = x \\
 y = S = 18 &\rightarrow 18+13 = 4 = E = x \\
 y = I = 8 &\rightarrow 8+13 = 21 = V = x \\
 y = S = 18 &\rightarrow 18+13 = 4 = E = x \\
 y = E = 4 &\rightarrow 4+13 = 17 = R = x \\
 y = N = 13 &\rightarrow 13+13 = 26 = \square = x \Rightarrow \text{First word} = \underline{\text{NEVER}}
 \end{aligned}$$

(b) Plaintext $E \rightarrow$ Ciphertext $T \Rightarrow 4 \rightarrow 19$ $x \rightarrow ax+b=y$
 Plaintext $S \rightarrow$ Ciphertext $V \Rightarrow 18 \rightarrow 21$ encryption function

$$\begin{aligned}
 x=4 &\rightarrow 4a+b=19 \Rightarrow 14a=2 \Rightarrow 7a=1 \Rightarrow a=7^{-1} \Rightarrow a=4 \\
 x=18 &\rightarrow 18a+b=21
 \end{aligned}$$

$$a=4 \Rightarrow 4(4)+b=19 \Rightarrow b=3 \Rightarrow (a,b) = \underline{(4,3)}$$

$$\begin{aligned}
 7 \times 4 &= 28 \\
 &= 1 \\
 &\text{mod } 27
 \end{aligned}$$

(c) When we study the security of a cryptosystem we assume that an adversary knows exactly which cryptosystem is being used but that they do not have the key being used. In this way, the overall security depends only on the mathematical properties of the cryptosystem and the plaintexts and keys used rather than our ability to keep the encryption/decryption algorithms safe (secret).

(d) $\phi(55) = \phi(5) \times \phi(11) = 4 \times 10 = 40 \Rightarrow 40$ possibilities for a
 55 possibilities for b
 $\Rightarrow 40 \times 55 = 2200$ possible keys

a can only be a number that is coprime to 55. Otherwise two letters would be encrypted to the same letter.

Q2 (a) Suppose that $|P| = |C| = |K|$. Then our cryptosystem is perfectly secure iff

- $\text{Prob}(K=k) = \frac{1}{|K|}$ for all $k \in K$
- $\forall x \in P, \forall y \in C, \exists! k \in K \text{ s.t. } ek(x) = y$

(b) Block ciphers operate on blocks of plaintext and ciphertext symbols, to produce blocks of ciphertext and plaintext symbols, respectively. Affine ciphers (e.g. DES (Data Encryption Standard)) are examples of block ciphers.

Stream ciphers operate on streams of symbols over an alphabet. Each plaintext symbol on the input stream is encrypted in a way that depends on the previous history of the stream.

- Plaintexts are streams $a_1, a_2, \dots$ over $\mathbb{Z}_n$

- Keys are seeds used to generate pseudo-random key streams $k_1, k_2, \dots$

- Encryption: $a_1, a_2, \dots \rightsquigarrow a_1 + k_1, a_2 + k_2, \dots$

- Decryption: $a_1, a_2, \dots \rightsquigarrow a_1 - k_1, a_2 - k_2, \dots$

Synchronous/asynchronous stream ciphers like A5/1 used in GSM (Global System for Mobile Communication) are examples of stream ciphers.

(c) A feedback linear shift register of length L is a feedback shift register of the form $f(s_{j-1}, \dots, s_{j-L}) = c_1 s_{j-1} + \dots + c_L s_{j-L}$ for fixed $c_1, \dots, c_L \in \mathbb{Z}_2$.

(d)

s_{j+2}	s_{j+1}	s_j	j
0	0	1	0
1	0	0	1
1	1	0	2
1	1	1	3
0	1	1	4
1	0	1	5
0	1	0	6
0	0	1	7

Initial sequence recovered $\Rightarrow$ period = 7

- Q3 (a) Symmetric cryptography consists of sets P, C, K of plaintexts, ciphertexts and keys and for each key $k \in K$, functions $ek: P \rightarrow C$ and $dk: C \rightarrow P$ with $dk(ek(x)) = x$ for all $x \in P$

Public key cryptography consists of sets P, C, K as before and functions $ek: P \rightarrow C$, $dk: C \rightarrow P$ for each $k \in K$ with the property that for each (public key) $e \in K$ there exists (a private key) $d \in K$ with $dk(ek(x)) = x$ for all $x \in P$

For symmetric cryptosystems, key distribution scales badly as the number of parties grows. For a group of $n \geq 2$ people to be able to communicate with each other securely, $\binom{n}{2}$ keys need to be exchanged. For public key cryptosystems, we only need n keys

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- (b) Alice picks two different, large primes p and q and computes their product $n = p \cdot q$. She then chooses a random integer e with $1 < e < (p-1)(q-1)$ and $\gcd(e, (p-1)(q-1)) = 1$.

Her public key is (n, e)

Next, using the Euclidean Algorithm she computes the unique integer d with $1 < d < (p-1)(q-1)$ and $d \cdot e \equiv 1 \pmod{(p-1)(q-1)}$. i.e. $d \in \mathbb{Z}_{(p-1)(q-1)}^*$ is the inverse of e modulo $(p-1)(q-1)$.

Her private key is (n, d)

(c) Shared key = g^{ab} . $g=2$, $a=6$, $b=3 \Rightarrow 2^{63} = 2^{48} \equiv \underline{19} \pmod{25}$

(d) $C = m^e \pmod{n}$. $m=23$, $e=17$, $n=77 \Rightarrow C = 23^{17} \pmod{77} \equiv \underline{67}$

Q4 (a) Euler's totient function ($\phi(n)$) counts all the numbers up to n that are co-prime to n .

Euler's totient theorem states that if you let $n \geq 1$ and $a \in \mathbb{Z}$, with $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

(b) Fermat's little theorem gives that if p is prime and $a \in \mathbb{Z}$ with $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$. Hence, if $a^{n-1} \not\equiv 1 \pmod{n}$ for $n \geq 1$ and $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$, then n is not prime. This gives us a method for proving that some integers are composite without constructing a divisor.

(c) A composite number $n \geq 4$ is a Carmichael number if $a^{n-1} \equiv 1 \pmod{n}$ for all $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. They are odd, square-free and divisible by at least three primes eg. 561. Fermat's primality test reports a Carmichael number is "probably prime" for any choice a base $a \in \mathbb{Z}_n^*$, making the test not reliable.

(H) Express the group structure as terms of coordinates. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be points on the elliptic curve $y^2 = x^3 + ax + b$ where $P \neq \theta \neq Q$ (θ is the identity element is the point at infinity) and $P \neq -Q$. Define

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & \text{if } P \neq Q \\ \frac{3x_1^2 + a}{2y_1} & \text{if } P = Q \end{cases}$$

Then $P + Q = (x_3, y_3)$ for $x_3 = \lambda^2 - x_1 - x_2$ and $y_3 = \lambda(x_1 - x_3) - y_1$.